

Chapter 2: Electrostatics Study Guide & Past Paper

Class 10 Physics • Chapter Reference & Exam Preparation

1. Core Theoretical Concepts

Electrostatics is the branch of physics that deals with the phenomena and properties of stationary or slow-moving electric charges.

- **Coulomb's Law:** Describes the electrostatic force of attraction or repulsion between two point charges. Formulated as: $F = k * (q_1 * q_2) / r^2$, where k is Coulomb's constant ($8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$).
- **Electric Field:** The region of space around a charged particle where another charge experiences an electrostatic force. Measured in Newtons per Coulomb (N/C).
- **Electric Potential:** The work done per unit charge in bringing a positive test charge from infinity to that point in space. Measured in Volts (V).
- **Capacitance:** The ability of a system to store an electric charge. Expressed as $C = Q/V$. The SI unit is the Farad (F).

2. Past Paper Mock Questions

Q1. [Multiple Choice - 2 Marks]

Two point charges q_1 and q_2 are placed at a distance r . If the distance between them is doubled, the electrostatic force F between them becomes:

(a) $2F$ (b) $4F$ (c) $F/2$ (d) $F/4$ [Answer: d]

Q2. [Descriptive Question - 5 Marks]

Define electric potential at a point. Derive the expression for the equivalent capacitance of three capacitors connected in parallel. Draw a neat circuit diagram to illustrate the connection.

3. Recommended Practical Assignments

- Investigate static charge generation using a glass rod rubbed with silk.
 - Determine the capacitance of a parallel plate capacitor using a standard dielectric sheet.
 - Calculate total charge stored in a bank of capacitors connected in series.
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